AFOLABI OLAWALE GOODLUCK

AI Engineer · Agentic Systems & LLM Applications · AWS Cloud Architecture

Lagos, Nigeria | lekanolawale477@gmail.com | +234 808 441 5996 | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI Engineer with 2+ years shipping production agentic AI, RAG, and LLM systems on AWS across banking, fintech, healthtech, and investment platforms. Built RubiesAI, an agentic banking assistant for a Qucoon bank client that executes real account actions — balance checks, transfers, purchases — directly from natural-language chat. Also built a 5-agent customer-support system handling 1,000+ daily conversations, a real-time streaming investment assistant with live web-search tool-use, and its companion support/admin platform. Deep hands-on experience across AWS Bedrock (Claude, Nova, Titan, GLM), serverless and containerised architecture (Lambda, ECS Fargate, SQS/SNS, IAM, VPC), and vector search (pgvector, Pinecone, S3 Vectors) — comfortable owning cloud architecture end to end, not just the model layer.

SELECTED IMPACT

- **RubiesAI:** built the agentic banking assistant that lets Rubies Bank customers check balances, buy, and transfer money by chat alone — Qucoon's flagship agentic-banking build.
- **1,000+ daily conversations** handled autonomously at 85–89% resolution across two production multi-agent systems (Motito, VestAI support).
- **70% / 60% faster:** cut documentation time 70% at GeroCare and response time 60% at Motito through real-time streaming and parallel agent execution.

TECHNICAL SKILLS

AI / LLM: AWS Bedrock (Claude Sonnet 4/4.5, Amazon Nova Pro, Titan, Z.AI GLM-5) · Multi-agent & agentic systems · Tool/function calling · RAG pipelines · Prompt engineering · Streaming inference

Vector & Data: PostgreSQL + pgvector · Pinecone · S3 Vectors · Redis / ElastiCache · Titan & OpenAI embeddings · Semantic + hybrid search · DynamoDB

Cloud & Solutions Architecture: AWS Lambda · ECS Fargate · SQS · SNS · EventBridge · S3 · IAM · VPC · CloudWatch · Application Load Balancer · SAM · ECR · Docker · GitHub Actions CI/CD

Backend & Real-Time: Python · FastAPI · Pydantic · Async/await · WebSocket streaming · REST APIs · Webhooks · WhatsApp Business / Meta Cloud API

WORK EXPERIENCE

Qucoon Limited — AI Engineer / Solutions Architect

Mar 2025 – Present

Lagos, Nigeria · Client-facing AI delivery, internal product development, and cloud architecture

RubiesAI — Agentic Banking Assistant — *Rubies Bank (Qucoon Client Engagement) · AWS Bedrock · Agentic Tool-Use · Production — Flagship Project*

- Built an agentic AI banking assistant for Rubies Bank that performs real banking actions — **checking account balances, buying, and transferring money** — directly from a customer's natural-language chat message, no menus or forms required.
- Designed a multi-tool agentic architecture on AWS Bedrock (Claude) where the model selects and calls the right banking tool per user intent (balance check, transfer, purchase); a backend execution layer validates and carries out the action against the bank's core systems.
- Built a dual-intent router that separates genuine transactional requests from ordinary conversation, with per-transaction business-rule checks and confirmation before any money movement executes.
- Delivered as Qucoon's flagship agentic-banking build — the reference tool-calling architecture other Qucoon fintech engagements now build from.

VestAI — Investment Assistant, Support & Disclosure Platform — *Qoovest · AWS ECS Fargate · Bedrock · WebSocket · Serverless*

- Own and built four production AI systems for Qoovest's NGX stock-investment platform: a real-time conversational investment assistant, its customer-support/admin platform, an automated regulatory-disclosure pipeline, and a market-news distribution pipeline.
- Built the conversational investment assistant end to end: a 6-stage agentic loop (Understand → Assess → Search → Reflect → Synthesize → Respond) on Claude Sonnet via Bedrock, a live web-search tool, and token-by-token WebSocket streaming to the frontend, backed by a dual-store memory architecture (Redis for hot context, DynamoDB for permanent history).
- Built the companion support/admin platform: a RAG-backed FAQ bot (Amazon Nova + Titan embeddings over S3 Vectors) with automatic human escalation, a first-claim-wins admin-assignment system, a live customer↔admin chat relay over Redis Pub/Sub,

integration into the separate ticketing backend admins work from, and automatic reminder alerts when an admin forgets to hand a conversation back to the bot.

- Built a serverless Lambda pipeline (Fetch → SQS → Process → Publish) that scrapes NGX disclosure filings twice daily, summarises each filing with Bedrock for retail investors, and routes it through a human-reviewer approval step before publishing — fully automated end to end.
- Built a market-news pipeline that fetches and deduplicates stories from multiple Nigerian finance outlets, groups same-event coverage and rates importance with Bedrock, then serves the right stories to the right investors through a dedicated news API — keeping retail users informed with zero manual curation.

Motito — Multi-Agent Customer Support System — *Fintech Commerce · Ghana · WhatsApp · Production*

- Built a 5-agent AI system (Intent, Conversation, Product, Support, Escalation) on AWS Bedrock handling **1,000+ daily queries** with **89% first-contact resolution**.
- Cut response time by **60%** by parallelising agent calls with asyncio.gather instead of running them sequentially.
- Designed a dual-store RAG pipeline (PostgreSQL + pgvector, Titan 512-dim embeddings, IVFFlat cosine index) covering both product search and the support knowledge base, using Claude 3.5 Sonnet to turn raw product JSON into search-oriented descriptions before embedding.
- Built a fully async, serverless ETL pipeline (API → SQS → Lambda) that decouples slow LLM/embedding work from the request path, with incremental per-portal Chatwoot sync and idempotent upserts — keeping the knowledge base current with zero manual intervention.
- Added Redis conversation memory (30-min TTL) and confidence-based escalation (0.0–1.0 threshold), keeping autonomous AI handling at **85%**.

Qucoon DocGen v2 — AI Document & Architecture-Diagram Platform — *FastAPI · AWS Bedrock (GLM-5) · Template Engine*

- Built a platform that turns a plain-English project brief into a fully branded .docx document or .pptx AWS architecture diagram in seconds — replacing a manual SOW/proposal-drafting process that used to take hours.
- Designed a template-driven generation engine (per-team Word/PowerPoint templates with placeholder + loop syntax) so new document types and diagram layouts ship with **zero code changes**.
- Added a document-upload path (PDF/DOCX/XLSX extraction → AI summarisation) that turns an uploaded brief directly into a generation-ready prompt.

GeroCare — AI Medical Copilot — *Healthcare · HIPAA-Aware · AWS Serverless*

- Designed a secure assistant with real-time voice-to-text transcription (Amazon Transcribe) and auto-generated SOAP reports via Claude Sonnet 4, using Amazon Nova Pro for fast real-time classification — cutting documentation time by **70%**.
- Built a read-only DB architecture — reads patient history from MySQL/RDS but writes only draft reports to S3 — zero risk to live clinical records.
- Added a two-pass validation pipeline: first pass drafts the report, second pass cross-checks every claim against the raw transcript and flags unsupported content.

FlowOS — Multi-Tenant WhatsApp Business SaaS — *Nigerian SMEs · AWS Lambda · Bedrock · Real-Time*

- Contribute to a multi-tenant AI commerce platform on AWS Lambda + RDS/pgvector + Bedrock (Claude Haiku 4.5 primary, Nova Pro fallback), including a hybrid semantic + keyword product-search layer and a 16-question dynamic AI onboarding flow that builds each business's own knowledge base.
- Verified real AI cost-per-reply unit economics to keep the model-routing strategy commercially sustainable at scale; built real-time WebSocket chat delivery and a role-based staff access system (order-manager / support roles) on top of the existing REST API.
- **Additional:** shipped Qorpy, a multi-tenant enterprise RAG FAQ platform on AWS Lambda with Pinecone namespace isolation — **1.75s median response time**, **3× faster** retrieval via sub-query decomposition, and CI/CD shipping updates in under 2 minutes.

PERSONAL PROJECTS

- **ConfidenceAI — AI Coaching App** (Gemini · Streamlit · Pydantic, 2024): real-time confidence scoring, multi-step reasoning, and goal-tracking analytics.
- **Smart Salary Predictor** (scikit-learn · Python · Pandas, 2024): end-to-end regression pipeline achieving $R^2 = 0.87$ on unseen data.

EDUCATION

Lagos State University *Aug 2021 – Jan 2025*

B.Sc. Mathematics & Education